



Efficient ternary comparator on CMOS technology

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ABSTRACT

Binary (base-2) comparators suffer from interconnect-complexity and associated power-delay, area, fan-in/out, reliability and parasitic-overhead. Ternary (base-3) counterparts can effectively reduce interconnect complexity due to more-data-carrying capability and hence aforesaid drawbacks. A new idea to compare 2-ternary-inputs is proposed using Double-Pass-Transistor-Logic realized on Normal-Process-Enhancement-type-MOS (NPEMOS-technology) without threshold modification. The design-strategy is first presented for 1-trit ternary-comparator, extended next to compare two n-trit ternary-inputs. A 4-trit ternary-comparator is designed to demonstrate the proposed idea and to evaluate and benchmark performance. Proposed design is optimized using BSIM4 device-model on 32 nm-CMOS-technology on 1.0V supply-rail at 27 °C temperature. Trit values “0”, “1” and “2” are coded with 0V, 0.5V and 1.0V respectively. Functionality of the design is validated through extensive T-Spice transient-simulations applying all possible dynamic test-patterns using PWL input-source of Tanner EDA V.16. Next, post-layout performance is analysed and compared. Process-Voltage-Temperature analysis is performed on: 1.1V supply-rail for fast-MOSFET, 1.0V supply-rail for typical-MOSFET and 0.9V supply-rail for slow-MOSFET in the temperature −40 °C–100 °C.

1. Introduction

Digital comparator is a decision-making building block in the data-path in many arithmetic and logic computations including address decoding in microprocessors/microcontrollers, process control applications, servo-motor control, remote sensing, computer display and Wireless Sensor Network (WSN) [1–8] and has been receiving attention from before [1]. Artificial Intelligence (AI), Internet-of-Thing (IOT) as well as Multi-Input-Multi-Output (MIMO) systems for next generation wireless communication need large scale of iterative comparison of digital data and hence high-speed digital comparison along with low-power dissipation is important for aforementioned applications. Some recent activities on binary comparator in open literature can be evidenced in Refs. [3–13].

Although binary (base-2) is the dominant number system adopted globally in the digital era, yet recent literature [14–20] reveals binary designs are losing relevance especially in large-input digital processing in digital systems on account of growing interconnect complexity, associated power-delay overhead, reliability-degradation (due to generated localized heating or hot-spot) apart from fan-in/fan-out issues. Thus for identical discrete or digital levels in a comparator, a ternary (base-3)

design offers less circuit complexity by engaging lesser number of design blocks apart from interconnect overhead as compared to a binary-design for same number of levels as explored in Refs. [21,22]. This also significantly reduces the on-chip-real-estate and increases reliability by controlling hot-spot, fan-in/out and power-delay overheads. The advantage becomes more significant as the number of levels increase. Simple calculations would reveal for M-levels while a binary system would need $\frac{M}{\log_{10} 2}$ bits, a ternary (base-3) system would need $\frac{M}{\log_{10} 3}$ Trits (ternary digits), providing thereby ~160% reduction in interconnect overhead alone. Investigation on Multi-Valued-Logic (MVL) [23,24] revealed that ternary (base-3) number-system could be a potential alternative to replace conventional binary number-system in order to deal with possible drawbacks associated with binary-based processing of digital data. More information carrying capability offers reduced I/O pins in case of ternary that in turn reduces interconnect complexity as compared to binary system. Also base-3 (ternary) is the closest to natural base “e” (“e” = 2.718) among all multi-valued bases and hence offers reduced digital error as compared to other MVL [14] system. Yet, despite all these proven advantages, circuit designers continue to ignore ternary-based designs often due to a mind-set problem, and continue with binary designs in bulk of the digital blocks.

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Some of the recent activities on Ternary Comparators are as follows: In 2009, X.Zeng et al. [25] applied Multi-Valued Switch-Signal Theory in order to compare magnitude of two parallel ternary input in ripple-carry manner. However, the idea failed to provide high-speed comparison due to dependency of succeeding stages on output of preceding stage. The input data must stay stable until the comparison process is over. Next, in 2011, R. Z. Zadeh et al. [26] proposed a ternary comparator based on ternary Quantum-Full-Adder-Cells (QFAC). Cascading of n-QFAC are employed to generate comparative results of two n-Qutrit Ternary Numbers. Once again this is a serial structure and suffers like [25]. C.Vudadha et al. in June 2012 [27] proposed Ternary Comparator using CNTFET by combining both binary and ternary logic cells through novel grouping-strategy. In 2018, N. S. Soliman et al. [28] proposed Memristor-CNTFET based ternary comparator design. Most recently in 2020, S. Ali Hosseini et al. [2] proposed novel Multi-Valued-Logic Comparator which is once again serial in architecture and offers demerits of [25,26]. The comparators proposed in Refs. [2,25–28] exploit the threshold variation of the more expensive CNTFETs to achieve ternary operation. The circuits proposed therein suffer from reliability and stability issues especially with thermal variation. The large fabrication cost and complexity along with short life span of CNTFET as compared to conventional MOSFET makes aforesaid alternatives [2, 25–28] further unattractive.

Even though there have been certain advantages in working with ternary logic and ternary algebra yet conventionally most circuit designers still work with binary logic. However, if we consider interconnect overhead and associated baggage for instance hotspot generation, parasitic capacitance or resistance and the delay associated with it, the ternary system most often outperform its binary counterparts particularly in complex data converters and processors. This has been the motivation behind exploring new design ideas on comparators using ternary logic and ternary system. Present study deals with a new design-idea for ternary (base-3) comparator using conventional Enhancement-Type-MOSFET device-technology in normal process condition. Leaf cells for the proposed work have been designed using Double Pass-transistor Logic (DPL) [29,30]. The design idea has been explained with 1-trit ternary comparator first and then extended (generalized) to compare two n-trit ternary input. A 4-trit ternary-comparator-design based on the proposed strategy has been demonstrated to evaluate the performance and optimized on 32 nm CMOS technology using BSIM4 device model at 27 °C temperature on 1.0V supply rail. Ternary digits “0”, “1” and “2” are represented by 0V, 0.5V and 1.0V respectively. Extensive T-Spice transient simulations with all possible custom test patterns have been carried out to validate the functionality of the proposed comparator. Next, the physical design of proposed 4-trit ternary comparator is done and the post layout performance with extracted parasitic is evaluated. A benchmarking comparative study in terms of speed-power efficiency with most recent competitive designs is presented. Finally, PVT analyses with fast, typical and slow MOS-models considering $\pm 10\%$ supply variation from nominal value in ambient temperature range of -40 °C to $+100$ °C are performed and the results are recorded.

The rest of the paper is organized as follows: Section-II explains the proposed strategy to compare two ternary inputs and presents the design architecture. The circuit design, transient response, post-layout evaluation and benchmarking comparison of the proposed comparator with some recent candidate designs are presented in Section-III. Section-IV concludes the work.

2. Design architecture of proposed comparator

We now present the design-strategy of the proposed ternary-comparator circuit. In this endeavor the 1-trit ternary comparator is explained first. The 1-trit ternary comparator is then applied to design an n-trit ternary comparator in scalable fashion. The top level block diagram of 1-trit ternary comparator circuit is shown in Fig. 1. Two 1-trit ternary inputs are denoted by “ A_T ” and “ B_T ” where the input “ B_T ” is compared

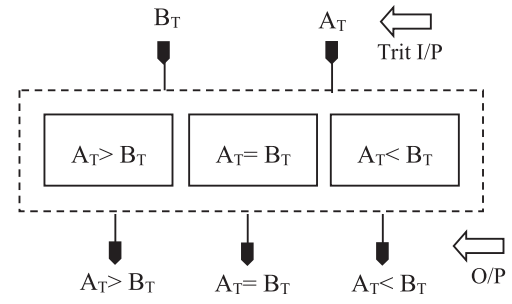


Fig. 1. 1-trit ternary comparator.

Table 1
I/O Relations for 1-trit ternary comparator.

Sl. No.	Input		Output		
	A_T	B_T	$A_T > B_T$	$A_T = B_T$	$A_T < B_T$
1.	0	0	0	2	0
2.	0	1	0	0	2
3.	0	2	0	0	2
4.	1	0	2	0	0
5.	1	1	0	2	0
6.	1	2	0	0	2
7.	2	0	2	0	0
8.	2	1	2	0	0
9.	2	2	0	2	0

with the input “ A_T ”. Three outputs, “ $A_T > B_T$ ”, “ $A_T = B_T$ ” and “ $A_T < B_T$ ” in Fig. 1 denote the three outcomes: “ A_T greater than B_T ”, “ A_T equal to B_T ” and “ A_T less than B_T ” respectively. The I/O relation for Fig. 1 is presented in Table 1. As per relative trit values of “ A_T ” and “ B_T ”, the corresponding outcome will be denoted by Ternary logic “2” and the other outcomes will be denoted by Ternary logic “0”.

Since ternary deals with three signaling levels, it requires three different inverters namely Positive Ternary Inverter (PTI), Negative Ternary Inverter (NTI) and Simple Ternary Inverter (STI). The I/O Relationship for all three ternary-inverters is shown in Table 2. Here in Table 2 the ternary input to the inverter is represented with “ A_T ” and the corresponding PTI, NTI and STI output, by PTI (A_T), NTI (A_T) and STI (A_T) respectively. $\overline{\text{PTI}}(A_T)$ and $\overline{\text{NTI}}(A_T)$ in Table 2 are obtained by applying STI on PTI (A_T) and NTI (A_T) outputs respectively. The circuit that detects trit value “1” at the input terminal is named as Middle Ternary Inverter (MTI) in the present work and the I/O relation for the proposed Middle-Ternary-Inverter (MTI) is also shown in Table 2. For ternary input $A_T = “1”$ the MTI output is either “2” or “0”. All three ternary inverters (PTI, NTI and STI) along with MTI are integral parts of proposed ternary comparator.

The logic diagram of proposed MTI to detect trit value “1” at input is shown in Fig. 2 and the corresponding circuit operation is summarized in Table 3. For further elaboration first consider ternary input “ $A_T = “0”$ ”, then the NTI (A_T) and $\overline{\text{PTI}}(A_T)$ outputs will be ternary logic “2” and at ternary logic “0” respectively. Since NTI (A_T) and $\overline{\text{PTI}}(A_T)$ offers two levels of signaling (i.e., either “0” or “2”), the binary NOR-gate has been used in the proposed design to generate the final MTI output which in this example is “0”. Following Similar operation the MTI (A_T) output for ternary input “ $A_T = “1”$ ” and “ $A_T = “2”$ ” will become “2” and “0”

Table 2
PTI, NTI, STI and MTI I/O relation.

Ternary input “ A_T ”	PTI (A_T)	PTI (A_T)	NTI (A_T)	NTI (A_T)	STI (A_T)	MTI (A_T)	MTI(A_T)
0	2	0	2	0	2	0	2
1	2	0	0	2	1	2	0
2	0	2	0	2	0	0	2

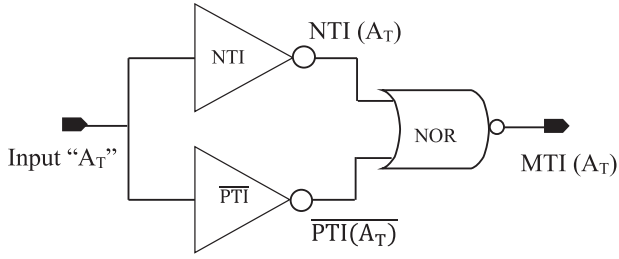


Fig. 2. Proposed middle ternary inverter (MTI).

respectively.

For circuit construction of proposed 1-trit ternary comparator first consider the condition $A_T > B_T$. The associated K-map is shown in Table 4 where, as per inherent pass characteristics of Enhancement-type NMOS and PMOS devices the optimized K-map grouping has been done and presented. The K-map grouping strategy is presented in Ref. [31] by A. Saha et al., in 2020, which is avoided here for brevity. The circuit detail for $A_T > B_T$ from K-map (Table 4) is elaborated below.

Part-I ($A_TB_T=00$): First consider the condition $A_T = 0$ and $B_T = 0$ in Table 4, for which " $A_T > B_T$ " output is 0. As per pass characteristics the NMOS transistor is used to pass "0" (0V) to the output. As a ternary (base-3) variable the input A_T and B_T have three possibilities "0", "1" and "2". For ternary input $A_T = "0", "1"$ and "2" the corresponding NTI (A_T) becomes "2", "0" and "0" respectively. Hence, an NMOS transistor with NTI (A_T) as gate input selects row-1 of K-map for " $A_T > B_T$ " (Table 4). Similarly, NTI (B_T) as gate input of an NMOS selects column-1 ($B_T = 0$) of the K-map (Table 4). Hence, Series combination of two NMOS transistor with gate input NTI (A_T) and NTI (B_T) has been applied here to select cross section of row-1 and column-1 (output for $A_T = 0$ and $B_T = 0$) of the K-map (Table 4). The complete circuit for the function $A_T > B_T$ is shown in Fig. 3. The logic-chain for $A_TB_T = 00$ is denoted by "Part-I" in Fig. 3.

Part-II ($A_TB_T=01, 02, 11$ and 12): Following aforesaid concept the circuit topologies for $A_TB_T = 01, 02, 11$ and 12 groups of output in K-map have been constructed by connecting two NMOS transistors in series with gate input as PTI (A_T) and NTI(B_T), and passing "0" (0V) to the output based on the status of gate input signal. For $A_T = "0", "1"$ or "2", the PTI (A_T) is "2", "2" or "0" respectively. Hence NMOS with PTI (A_T) as gate input selects row-1 and row-2 of the K-map (Table 3). Likewise the NMOS

Table 3
MTI circuit operation.

Input " A_T "	NTI (A_T)	PTI(A_T)	MTI (A_T)
0	2	0	0
1	0	0	2
2	0	2	0

with NTI(B_T) gate input selects column-2 and column-3 of Table 4. As a result, the series connection of them selects cross sections of row-1,2 and column -2,3. The corresponding logic chain is denoted by Part-II in Fig. 3.

Part-III ($A_TB_T=10$ and 20) and Part-IV ($A_TB_T=20$ and 21): As the outputs for Part-III and IV are at ternary logic "2" the PMOS transistor has been applied to generate the corresponding circuit topologies. Two series connected PMOS transistors controlling with NTI (A_T) and NTI(B_T) as gate-input selects cross sections of row-2,3 and column-1 of Table 4 and has been applied here for Part-III. Similarly for Part-IV, two series connected PMOS with gate-input PTI (A_T) and PTI(B_T) that selects cross sections of row-2 and column-1, 2 has been applied. The logic chain for Part-III and Part-IV are represented with dotted line in Fig. 3.

Part-V ($A_TB_T=22$): The cross section of row-3 and column-3 has been selected with two series connected NMOS transistors with gate inputs PTI(A_T) and PTI(B_T) respectively. For $A_T = "0", "1"$ or "2" the PTI(A_T) output is "0", "0" or "2" in that order and hence row-3 is selected when applied as a control input to the NMOS gate terminal. Again with PTI(B_T) as gate-input of an NMOS transistor selects column-3 of Table 4. Hence series connection of them serves the purpose. The logic construction for Part-V is shown in Fig. 3.

The aforesaid concept has been applied to construct circuit structure for " $A_T = B_T$ " and " $A_T < B_T$ " for proposed 1-trit ternary comparator design. The optimized K-map groupings are shown in Tables 5 and 6. The corresponding circuit diagram for " $A_T = B_T$ " and " $A_T < B_T$ " function is shown in Fig. 4 and in Fig. 5 respectively.

The signal flow structure for n-trit ternary comparator based on proposed strategy is shown in Fig. 6. Consider two n-trit ternary numbers as "A" ($A_{n-1} \dots A_1 A_0$) and "B" ($B_{n-1} \dots B_1 B_0$) where "B" is being compared with "A". The proposed 1-trit comparator has been used here first to compare each corresponding trit value (with equal weight/same

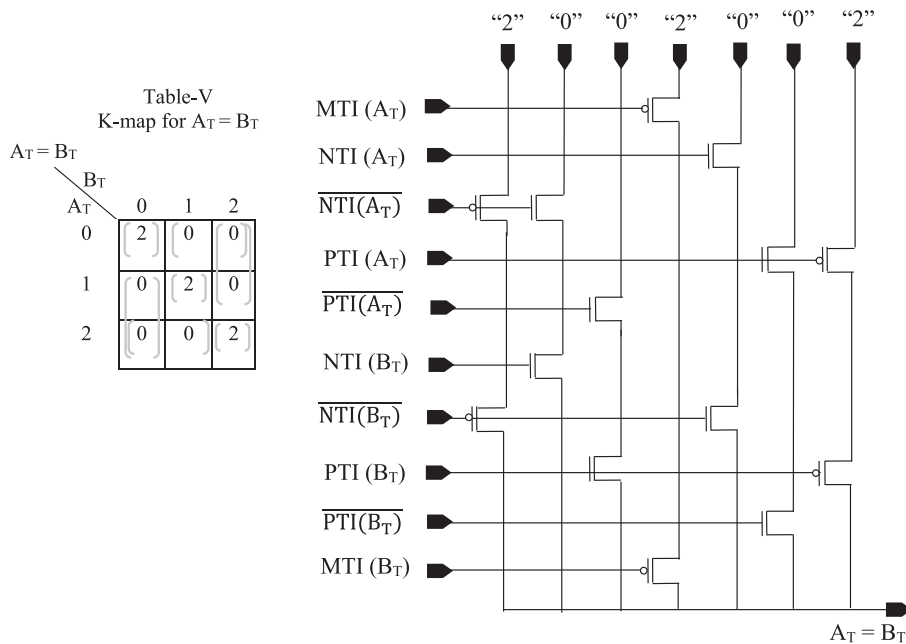
Fig. 3. Circuit diagram for $A_T > B_T$.

Table 4
K-map for $A_T > B_T$.

K-map for $A_T > B_T$

		B_T			
		0	1	2	
A_T	0	0	0	0	Row-1
	1	2	0	0	Row-2
	2	2	2	0	Row-3
		Col.-1	Col.-2	Col.-3	

Table 5
K-map for $A_T = B_T$.

K-map for $A_T = B_T$

		B_T			
		0	1	2	
A_T	0	2	0	0	
	1	0	2	0	
	2	0	0	2	

Table 6
K-map for $A_T < B_T$.

K-map for $A_T < B_T$

		B_T			
		0	1	2	
A_T	0	0	2	2	
	1	0	0	2	
	2	0	0	0	

position) of “A” and “B”. Since 1-trit comparator output offers two possible logic levels (i.e., either “0” or “2”), the binary (base-2) logic cells are utilized in intermediate processing stages to operate on trit-wise comparative results and generate the final n-trit-comparison output.

In present work a 4-trit ternary comparator has been constructed based on aforesaid strategy. The circuit topologies for intermediate processing stages of proposed 4-trit comparator are shown in Fig. 7. Here G_n , E_n and S_n represent “Greater”, “Equal” and “Smaller” outputs in the nth trit comparison (where “n” = 0, 1, 2, 3 for the 4-trit comparator) respectively in Fig. 7. The design of the proposed 4-trit ternary comparator along with transient validation, evaluation, comparative study with recent candidate designs and corner analysis report is given next in Section-III.

3. Circuit design, validation, evaluation and comparison

The proposed ternary comparator has been designed and optimized using BSIM4 model parameters on 32 nm CMOS technology at 27 °C temperature on 1.0V supply-rail using conventional E-MOS transistors without any threshold modification as explained in Section-II. 1-trit

ternary comparator based on the proposed idea has been designed first by representing trit values “0”, “1” and “2” by 0V, 0.5V and 1.0V respectively. The circuit optimization has been done by considering inherent pass and mobility characteristics for E-NMOS and E-PMOS devices. The NTI and PTI circuit for proposed work is shown in Fig. 8. The circuit operation for NTI and PTI circuit of Fig. 8 have been summarized in Table 7. The DPL binary-NOR circuit for proposed MTI (Fig. 2) has been designed following the concept of A. Saha et al. [29]. The designed circuit has been validated by investigating extensive T-Spice transient response for all possible test patterns at 27 °C temperature. The custom transient ternary-input to the comparator circuit have been applied through PWL (Piece-Wise-Linear) input source. Next the 1-trit comparator has been exploited to design the 4-trit ternary comparator on 32 nm CMOS technology on 1.0V supply rail at 27 °C temperature and also validated through extensive transient analysis for all possible test patterns with T-Spice simulation. Finally, the layout of proposed comparator has been done and is shown in Fig. 9.

After performing DRC (Design Rule Check) and investigating LVS (Layout Versus Schematic) of physical design the parasitic are extracted. Next the post-layout transient simulation applying extracted parasitic has been performed and is shown in Fig. 10. Here in transient simulation the custom input rise/fall time is set to 10ps. The PWL input source has been used to generate custom transient ternary input for the T-Spice simulation. Table 8 presents the evaluated characteristics from post layout simulation on 32 nm CMOS technology for proposed 4-trit ternary comparators. A total of 10,000 random test patterns have been applied dynamically to arrive at evaluated results of Table 8 (see Fig. 11).

As presented in Table 8, the core silicon area requirement for the proposed 4-trit ternary comparator is $3.51 \times 10^2 \mu\text{m}^2$ (Fig. 9). The physical area consumed for four $\times$ 1-trit comparators taken together, in the 4-trit comparator design is 48.56% of total silicon area. 35.49% of the total silicon area has been utilized for interconnections and power supply purpose. The power-supply and ground lines are aligned along the left and right side in the layout respectively. Whereas the input and output pins in the layout are placed along the top and bottom side respectively. 29.13% of total silicon area is utilized for the purpose of interconnections between four $\times$ 1-trit comparators and the intermediate processing stage the for 4-trit comparator. The propagation delay for the designed circuit has been measured by considering time delay between the 50% of input rail and 50% of output rail. The propagation delay for proposed 4-trit comparator is observed to be 27.03ps and dissipates 8.12 μW average power with Power-Delay-Product of $21.95 \times 10^{-5} \text{pJ}$.

The PVT analysis is also performed to investigate power-delay characteristics of proposed 4-trit ternary comparator with different Process Voltage and Temperature conditions and the result is recorded in Table 9. Here, in this process the fastest, typical and slowest MOS models have been used on supply-rails of 1.1V, 1.0V and 0.9V respectively. Speed-power performance has been measured for -40°C , 27°C and 100°C ambient temperature. Performance evaluation has been carried out with 10,000 sets of random dynamic test patterns on 32 nm CMOS technology. As per the PVT analysis report the proposed 4-trit comparator offers the worst PDP result for the supply-rail of 1.1V with fastest device at 100°C temperature and the best PDP result for 0.9V supply-rail with slowest MOSFET at -40°C temperature. Fastest 4-trit comparison occurs on 1.1V supply, Fast MOSFET at -40°C and takes 18.76ps time to compare two 4-trit numbers. Slowest circuit operation takes 39.66ps time when operated on 0.9V supply, Slow MOSFET at 100°C temperature. Proposed comparator dissipates maximum average power of 12.19 μW when operated on 1.1V supply, Fast MOS transistor at -40°C temperature and minimum average power of 5.86 μW , on 0.9V supply, Slow MOSFET at 100°C ambient temperature.

Notwithstanding significant advantages over its binary counterpart, the complexity of ternary logic circuit due to handling of 3-levels of signaling is a major hurdle in its way to replace its binary-counterparts. Although ternary-logic reduces interconnect complexity on one hand for processing same amount of data (as established by W. Alexander in

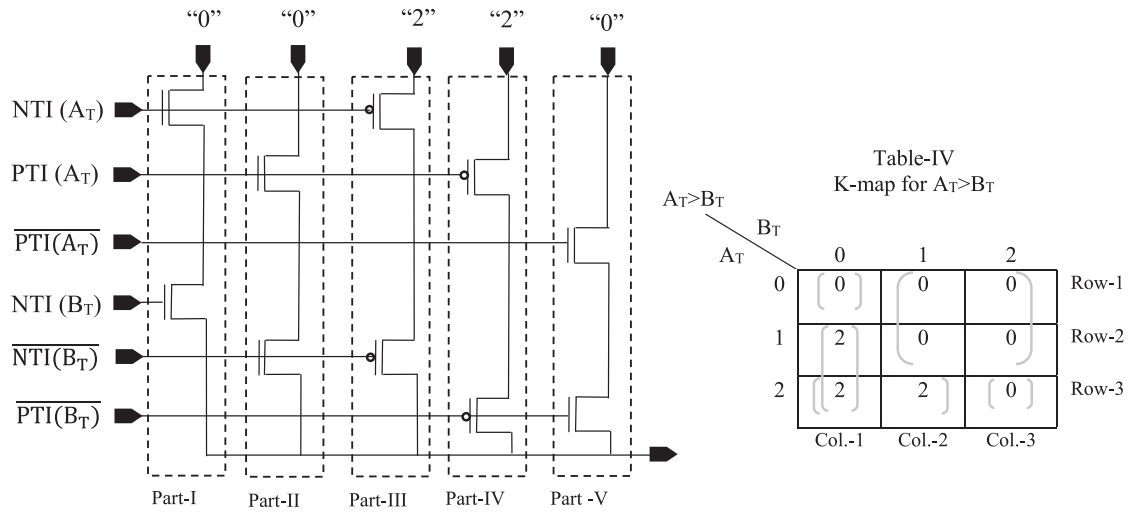
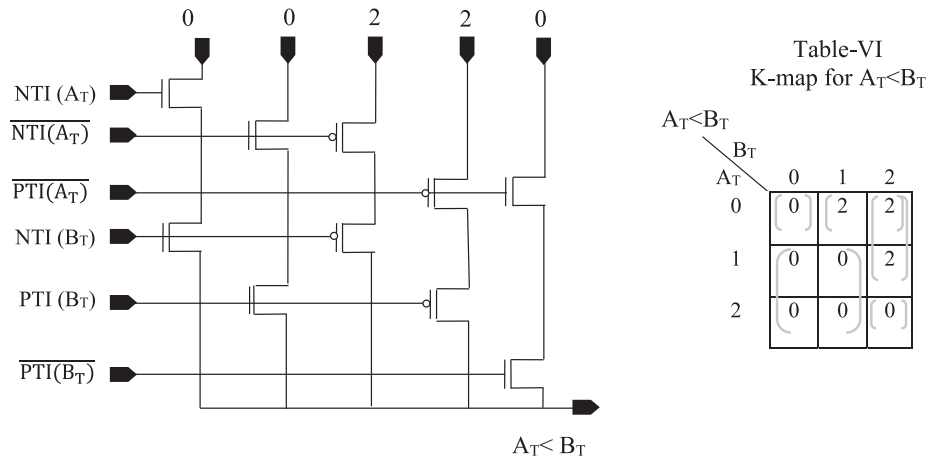
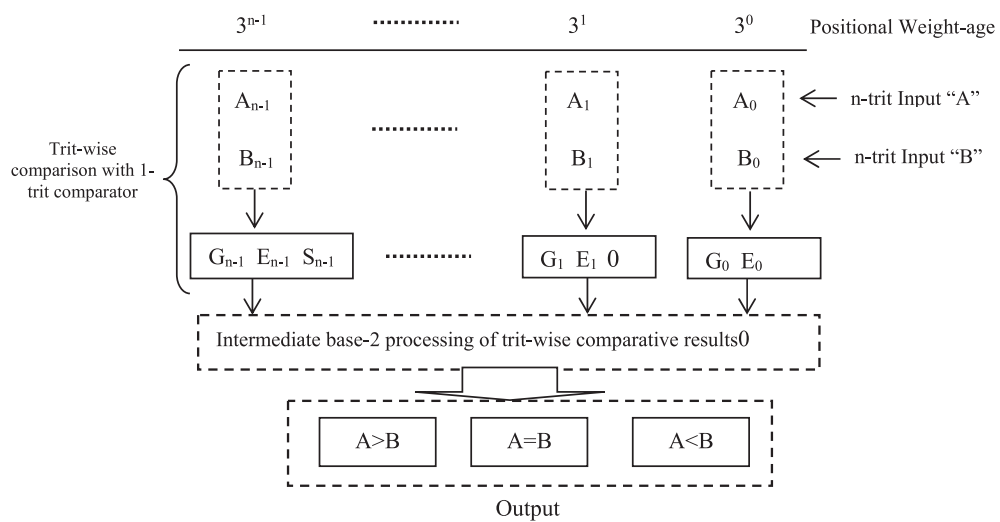
Fig. 4. Proposed circuit diagram for $A_T = B_T$.Fig. 5. Proposed circuit diagram for $A_T < B_T$.

Fig. 6. Proposed n-trit ternary comparator signal flow diagram.

1964 [21] as well as by S. L. Hurst in 1984 [22]), it increases logic complexity as compared to its binary counterpart. Clever design strategy however can overcome this hurdle to make the ternary-logic based

design attractive. Presented in this work is one such design strategy for ternary comparator, and thus has been favourably comparable in terms speed-power performance with two most recent binary comparators from

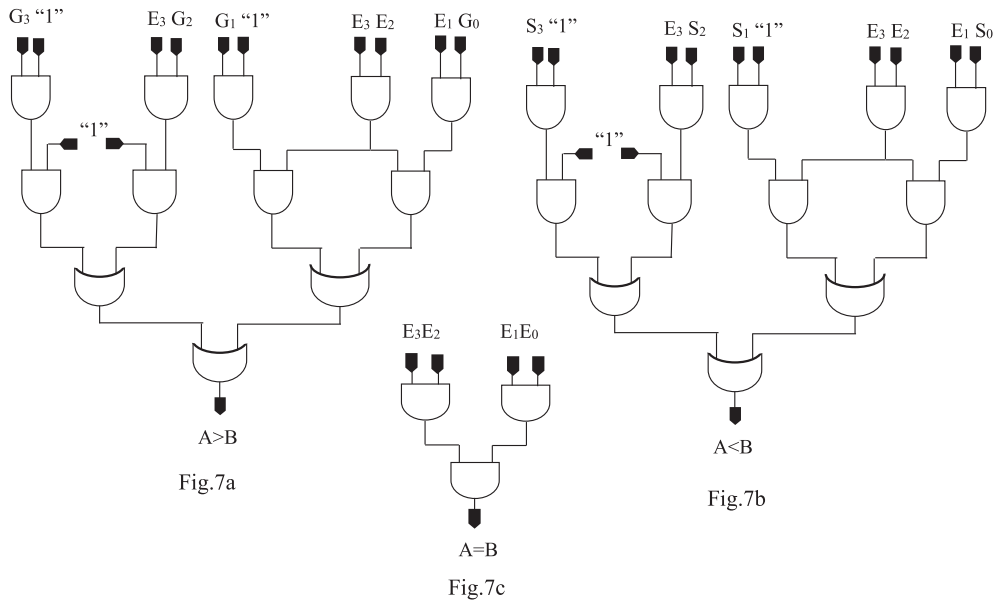


Fig. 7. Intermediate circuit to compare two 4-trit numbers “A” and “B” for Fig. 7a $A > B$ Fig. 7b $A < B$ and Fig. 7c $A = B$.

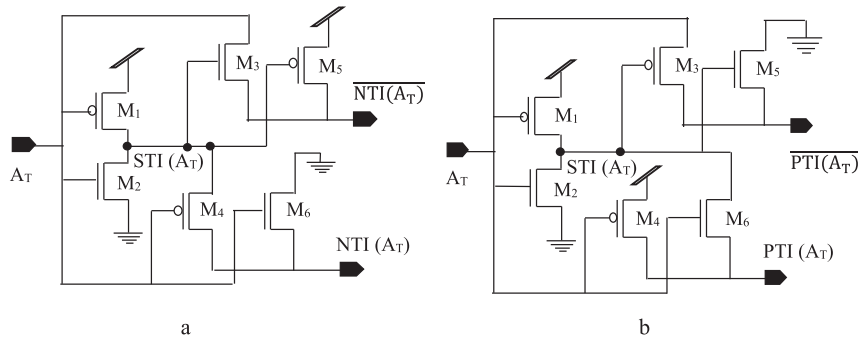


Fig. 8. Circuit Diagram for a). NTI and b). PTI.

Table 7

Circuit operation of NTI (Fig. 8a) and PTI (Fig. 8b).

Input A_T	Circuit Configuration: NTI (Fig. 8a)						Output		
	MOSFET Status								
	M_1	M_2	M_3	M_4	M_5	M_6	STI (A_T)	NTI (A_T)	$\overline{NTI(A_T)}$
“0”	ON	OFF	ON	ON	OFF	OFF	“2”	“2”	“0”
“1”	ON	ON	OFF	OFF	ON	ON	“1”	“0”	“2”
“2”	OFF	ON	OFF	OFF	ON	ON	“0”	“0”	“2”

Input A_T	Circuit Configuration: PTI (Fig. 8b)						Output		
	MOSFET Status								
	M_1	M_2	M_3	M_4	M_5	M_6	STI (A_T)	PTI (A_T)	$\overline{PTI(A_T)}$
“0”	ON	OFF	OFF	ON	ON	OFF	“2”	“2”	“0”
“1”	ON	ON	OFF	ON	ON	OFF	“1”	“2”	“0”
“2”	OFF	ON	ON	OFF	OFF	ON	“0”	“0”	“2”

open literature for the processing of identical amount of data. The comparative results are presented in Table 10. In order to maintain fairness in comparative study the comparators are compared on the same technology platform (32-nm Standard CMOS) and on 1.0V supply-rail at 27 °C temperature. As per the analyses shown in Table 10, the proposed design achieves nearly 58.25% reduction in power-dissipation, 24.41% reduction in delay and 68.44% reduction in PDP as compared to the

next-best candidate-design proposed by M. Rahimi et al., in 2019 [3]. Analyses and comparative study thus establishes the proposed idea based on ternary system, as an effective design of choice to replace binary comparators.

Next, the proposed 4-trit ternary comparator is compared with some recent candidate-ternary comparator-designs in terms of Power-dissipation, Propagation-delay and Power-Delay-Product (PDP). In

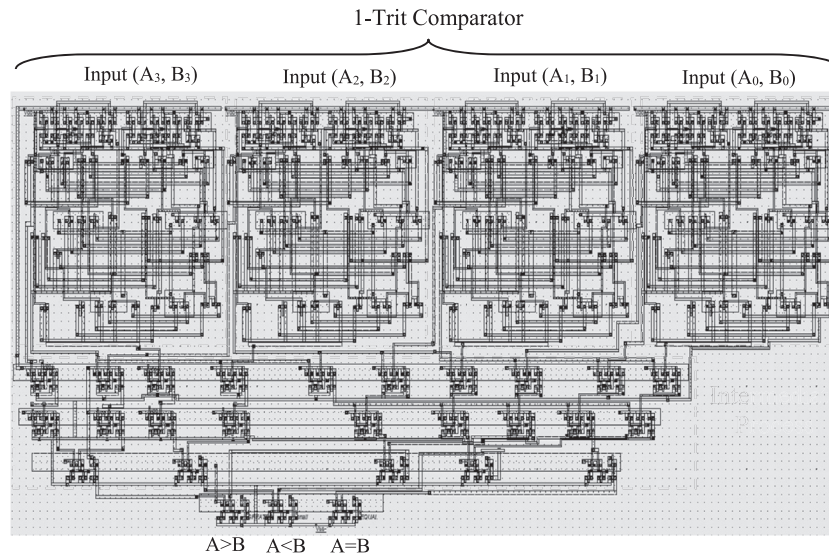


Fig. 9. 4-trit comparator.

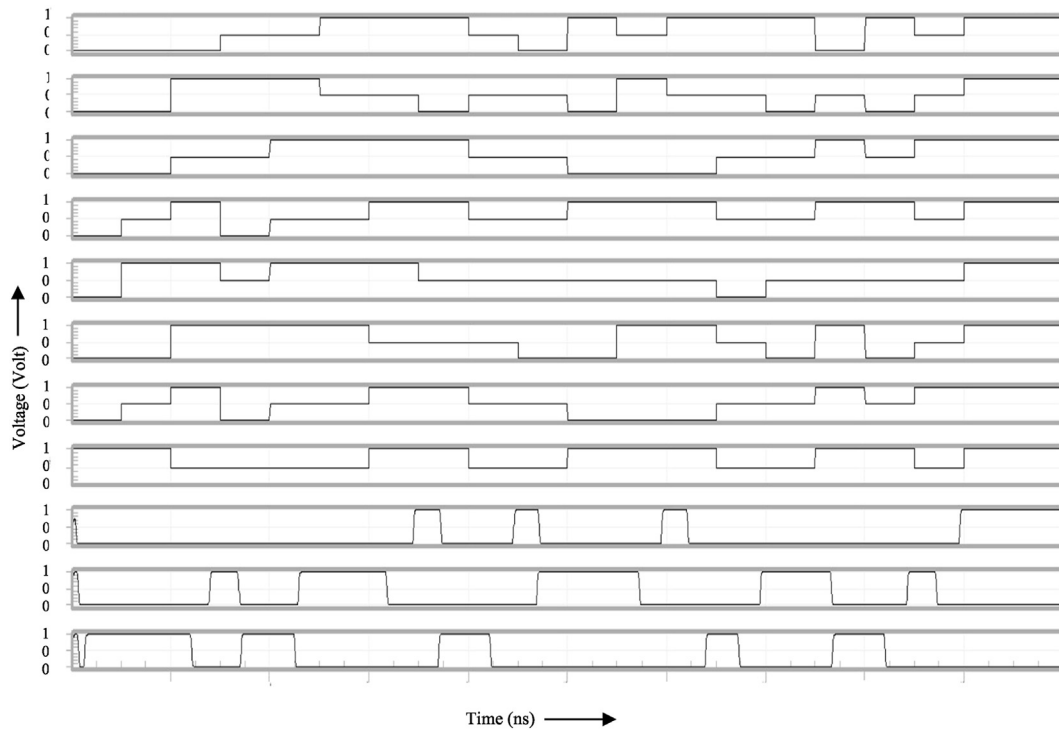


Fig. 10. Transient response of proposed 4-trit ternary comparator.

order to keep fairness in the comparative study all the ternary comparators are compared after realizing them on 32-nm standard CMOS technology with 1V supply-rail and identical operating condition by applying 10,000 random test pattern at 27 °C temperature. Comparing standard-CMOS based comparator with CNTFET counterpart is unfair. In this context the CNTFET based candidate available in open literature [2, 27 and [32] are evaluated applying multi-threshold MOSFET device as per the concept presented in Refs. [23,33]. From the comparative study presented in Table 11, the ternary comparator proposed by S. A Hosseini et al., in 2019 [2] and X. Zeng et al., in 2009 [25] offer better power response as compared to the proposed one only due to use of serial architecture. However, both underperform and therefore lose in terms of delay as well as Power-Delay-Product (PDP) as compared to the proposed

design. The proposed ternary comparator offers 84.28% and 79.59% overall delay reduction over the two best candidate-designs proposed in Refs. [2,25] respectively. The proposed design also reports 48.71% and 55.49% PDP-reduction as compared to the ternary comparators proposed in Refs. [2,25] respectively. The study thus once again reaffirms that the proposed idea can be an effective alternative and a design of choice to replace the binary candidates.

4. Conclusions

Present endeavor proposes a novel speed-power efficient scalable ternary comparator using Double Pass-transistor Logic (DPL) realized with conventional E-MOSFET device-technology. A 1-trit ternary

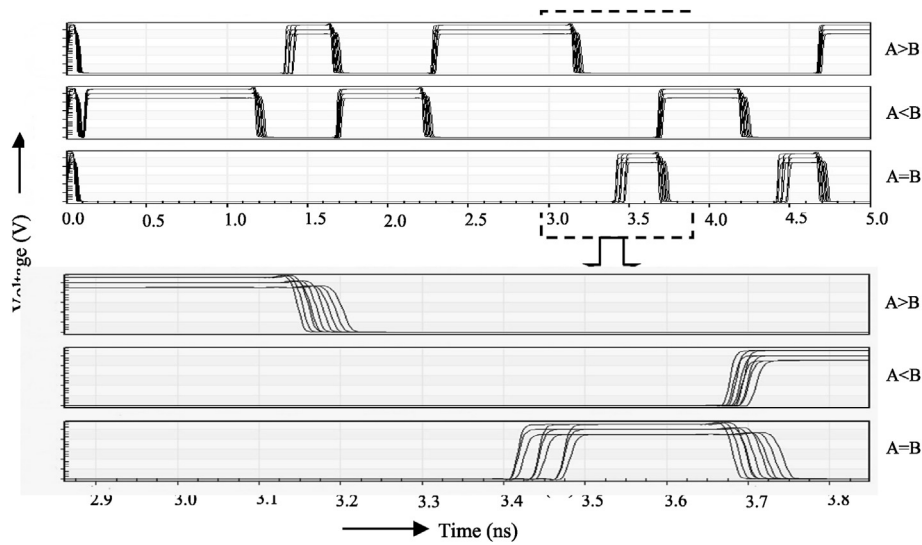


Fig. 11. PVT analysis response.

Table 8

Evaluated performance of proposed 4-trit ternary comparator.

Design	Silicon Area (μm^2)	Parameters			Tech.	Temp.	Supply-rail	"Trit" Value
4-trit Comparator	3.51×10^2	Delay (ps)	Avg. Power (μW)	PDP (pJ)	32 nm CMOS Process	27 °C	1.0V	"0" = 0.0V "1" = 0.5V "2" = 1.0V
		27.03	8.12	21.95×10^{-5}				

Table 9

PVT analysis speed-power report for proposed 4-trit ternary comparator.

Sl. No.	Parameter	PVT Analysis Response								
		Supply 1.1 V, Fastest MOSFET			Supply 1.0 V, Typical MOSFET			Supply 0.9 V, Slowest MOSFET		
		−40 °C	27 °C	100 °C	−40 °C	27 °C	100 °C	−40 °C	27 °C	100 °C
1	WC Delay (ps)	18.76	23.74	27.89	21.38	27.03	31.92	25.81	33.37	39.66
2	Avg. Power (μW)	12.19	10.25	8.91	9.17	8.12	7.29	6.94	6.35	5.86
3	PDP (PJ)	22.87×10^{-5}	24.33×10^{-5}	24.85×10^{-5}	19.61×10^{-5}	21.95×10^{-5}	23.29×10^{-5}	17.92×10^{-5}	21.19×10^{-5}	23.26×10^{-5}
4	Norm. PDP	0.92	0.98	1	0.79	0.88	0.94	0.72	0.85	0.93

Table 10

Performance benchmarking of proposed ternary comparator with two most recent binary comparators for equal amount of data processing.

Design	Year	Radix	Power (μW)	Delay (ps)	PDP (pJ)
R. Gupta et al. [10]	2017	Binary	13.79	65.21	89.92×10^{-5}
M. Rahimi et al. [3]	2019	Binary	19.45	35.76	69.55×10^{-5}
Proposed	Present	Ternary	8.12	27.03	21.95×10^{-5}

comparator is designed first and then scaled up to realize a 4-trit ternary comparator for comparison of two 4-trit ternary numbers using the same interconnect block. Thus the proposed design is general and upward scalable on a modular basis – an aspect that has increased its application-potential manifold.

All the design optimizations are performed on 32 nm CMOS technology on 1.0V supply rail at 27 °C temperature using Tanner EDA V.16.

In present work the ternary-digit "0", "1" and "2" are represented by 0V, 0.5V and 1.0V respectively. The functionality of proposed comparator has been validated through extensive T-Spice transient simulations with all possible sets of dynamic random ternary test-patterns using PWL (Piece Wise Linear) source. Next, the layout of proposed comparator has been done. After performing DRC, LVS-check and parasitic extraction the post layout simulation has been carried out to retrieve actual circuit response. The corner analysis with PVT variation has been performed applying Fast MOSFET on 1.1V supply rail, Typical MOSFET on 1.0V supply rail and Slow MOSFET on 0.9V supply rail to find power-delay characteristics of proposed 4-trit ternary comparator within temperature range of −40 °C–100 °C.

In the net analysis the benchmarking exercise conducted, establish the superiority of the proposed design in terms of speed-power response as compared most recent competitive candidate design to prove the efficiency-point. It thereby conclusively places the proposed comparator as a strong contender and a design of choice for speed-power efficient ternary data comparison.

Table 11
Performance benchmarking with some most recent competitive design.

Tech: 32 nm CMOS				
Temp.: 27 °C				
Design	Trit value	Power (μW)	Delay (ps)	PDP (J)
S.A. Hosseini et al. [2]	"0" = 0.0V	2.48	171.98	4.27×10^{-16}
S. L. Murotiya et al. [32]	"1" = 0.5V	12.15	62.89	7.64×10^{-16}
Vudadha et al. [27]	"2" = 1.0V	11.29	73.67	8.32×10^{-16}
X.Zeng et al. [25]		3.71	132.48	4.92×10^{-16}
Proposed		8.12	27.03	2.19×10^{-16}

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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